# The Japanese Animation and Home Video Game Industries: Locational Patterns, Labor Markets, and Inter-firm Relationships

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## I Introduction

The Hollywood's motion picture industry is probably the world's most famous cultural industry. Although it has long history and tremendous influence over foreign countries, local cultural industries still survive in individual countries. Indeed some of them have grown into regional and even global cultural industries. For instance, Japanese cultural products have spread to all over the world, and some of them change to the movies remade by the Hollywood. These cultural products are animation, cartoons, and home video games.[1)]

Usually we believe this phenomenon arises from the circumstances unique to Japan.[2)] This is why the value of cultural products is literally rooted in geographical variations between countries and regions. On occasions, regional agglomerations reinforce the competitive advantage of cultural products in both production and distribution.[3)] In discussing the competitiveness regional (or national) cultural industries some investigators concentrate only on agglomeration economies affecting the production sector;[4)] others argue that the distribution sector is more sensitive to the influence of agglomeration economies than the production sector,[5)] or alternatively contend that agglomeration economies are insignificant.[6)]

Surprisingly, we do not know the details of the individual cultural industries sufficiently to determine if these notions are true or false. Recently, studies about cultural industries have been

1) Japanese animation and cartoons are particularly known as *Anime* and *Manga*, respectively, in the world.

2) For instance, Aoyama, Y. and Izushi, H., 'Hardware gimmick or cultural innovation? Technological, cultural, and social foundations of the Japanese video game industry,' *Research Policy*, 32, 2003, pp. 423–444. They pay attention to the existence of a cartoonist, *TEZUKA Osamu*, who is regarded as the greatest person making cartoons a kind of arts and a starter of the animation TV series in Japan.

3) Scott, A. J., *The cultural economy of cities*, Sage, 2000, pp. 2–15.

4) (1) Storper, M. and Christopherson, S., 'Flexible specialization and regional industrial agglomerations: the case of the U. S. motion picture industry,' *Annals of the Association of American Geographers*, 77, 1987, pp. 104–117, (2) Christopherson, S., 'Project work in context: regulatory change and the new geography of media,' *Environment and Planning A*, 34, 2002, pp. 2003–2015, (3) Coe, N, M., 'The view from out West: embeddedness, inter-personal relations and the development of an indigenous film industry in Vancouver,' *Geoforum*, 31, 2000, pp. 391–407, (4) Mossig, I., 'The founding of new firms and efficient decision-making structures in localized production networks. The example of television production in the Cologne Media Cluster (Germany),' *Paper prepared for the 42nd European Regional Science Association (ERSA)*, Dortmund (Germany), 2002, pp. 1–21.

5) (1) Scott, A. J., 'A new map of Hollywood: the production and distribution of American motion pictures,' *Regional Studies*, 36, 2002, pp. 957–975, (2) Norcliffe, G. and Rendace, O., 'New geographies of comic book production in North America: the new artisan, distancing, and the periodic social economy,' *Economic Geography*, 79, 2003, pp. 241–263.

6) Aksoy, A. and Robins, K., 'Hollywood for the 21st century: global competition for critical mass in image markets,' *Cambridge Journal of Economics*, 16, 1992, pp. 1–22.

surely increasing both in quality and quantity. Yet, there is still a tendency to over-generalize. Rarely do we shed light on differences between cultural industries. Such behavior misleads argument about their true nature. Firstly, there are a very wide variety of cultural industries, which lead to diversity of their backgrounds: industrial history, market size, governmental assistance, etc. Secondly, there are either tangible manufactured goods or intangible services depending on a type of cultural industries.[7] Thus, it might be, for example, difficult to find common points in dispute between classic ballet and motion picture.[8]

In this paper, we compare two of the most widely-known contemporary Japanese cultural industries: the animation and home video game industries (game industry). Then we examine the main causes that underpin these differences. This attempt will enable us to introduce appropriate points of view sorting out the mess of argument.

The original data used in this paper was collected in 2000 on the animation industry and two years later on the game industry.[9]

## II Brief histories and business systems

### (1) The depressed new and the booming old

Japanese animation has had relatively long history. Although the industry's origins can be traced to 1917, it was not industrialized until 1953 when *Toei Animation* (*then Toei Doga*) was established. *Toei Animation* was a legitimate successor because it drew upon animators who had worked on projects prior to World War II,[10] and has continued to be the country's largest enterprise. In turn, *Toei Animation* was the sole source of trained animators. *Mushi Production*[11] was one of the beneficiaries of such labors, and produced country's first televised animation series.[12] The televised animation series have defined main characteristics of the Japanese animation industry, as argued below.

In contrast, the home video game industry is much younger than its animation counterpart. The industry was rooted in arcade games established in the United States that were developed from the late 1970s with computer technology advancing. In 1982 the industry was annihilated in the United States following the market crash — the so-called "Atari Shock". In 1983, Nintendo commenced selling the Family Computer[13] in Japan and established the foundation of the con-

7) Scott, A. J., 'The cultural economy of cities,' *International Journal of Urban and Regional Research*, 32, 1997, pp. 323–339.

8) Nevertheless, Caves points out six common property of the cultural industries: "nobody knows", "art for art's sake", "motley crew", "infinite variety", "time flies", and "ars longa." Caves, R.E., *Creative industries*, Harvard University Press 2000.

9) From July to December in 2000, mail questionnaire (71 of 287 firms) survey and interviews (33 firms) with the animation firms were carried out by the author. And also the same kind of survey intended for the game industry was conducted — respondents are 54 of 441 firms and interviewees are 50 firms — from May to November in 2002. To know more information, see (1) Hanzawa, S., 'Tokyo niokeru animehsyon sangyou syuuseki no kouzou to henyou,' *Annals of the Japan Association of Economic Geographers*, 47, 2001, pp. 288–302 (JE), (2) Hanzawa, S., 'Kateiyou bideo gemu sangyou no bungyou keitai to kuukan tokusei,' (JE) (submitted).

10) Yamaguchi, K. and Watanabe, Y., *Nihon animehsyon eiga shi*, Yubunsha, 1977. (J)

11) It was founded by *TEZUKA Osamu* — see footnote 2) — in 1962, and bankrupted in 1973.

12) Exactly to say, *Otogi Puro* produced 4 minutes animation program in 1961, and Mushi Production is a starter of half-hour animation program, which is the standard style in Japan.

13) It is called NES (Nintendo Entertainment System) in the U.S.

temporary game industry. Nintendo continued to play a leading role until Sony Computer Entertainment Inc. (SCE) took over its supremacy by introducing Playstation in mid-1990s.

Despite its relatively short history the home video game industry has experienced a deep recession in Japan, whereas the animation industry enjoys strong business success (Figure 1). The home video game market comprises both hardware (console) and software. Although the release of new generation consoles has increased hardware sales, software sales have continued to decline since 1997. While the causes of this recession are still obscure, many reasons are advanced to account for the decline in software sales. The reasons include: the diversification of the entertainment industry accompanying the diffusion of the cellular phone; a sharp decline in the birthrate; and increasing complications and difficulties in the content of games software. Some of these adverse conditions also affect the animation industry, and its market size is much smaller than that of game industry at that; however its economic performance is in better shape because of the growing global reputation of Japanese animation.[14] This reputation has enabled the Japanese animation industry to extend its market and, in the process, is attracting an increasingly number of entrants from other business sectors.

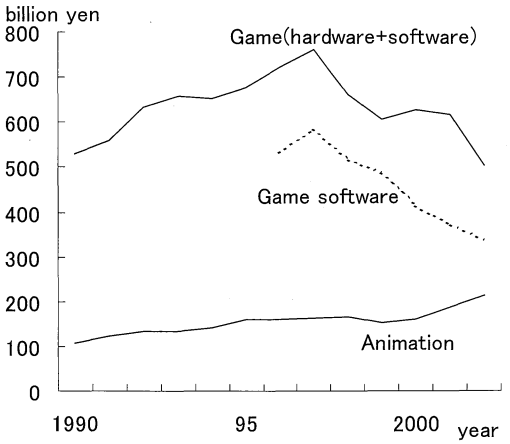



**Figure 1.** Market size of home video game and animation indusry in Japan

Note) Concerning game indusry, the data is based on *leisure hakusyo* until 1995 and CESA gemu hakusyo from 1996

Source:

1) Dentsu Souken., eds., *Jyoushou media hakusyo 2004*, Diamond Inc, 2003. (J)
2) *Leisure hakusyo* of each year published by Japan productivity center for socio-economic development. (J)
3) CESA game hakusyo of each year published by CESA. (J)

### (2) How does distribution system operate?

We sometimes focus on the production segment in discussing the cultural industry.[15] However, the distribution segment is critical due to the uncertainty of demand for aesthetic products.[16] Differences between production and distribution activities stem from their dissimilar prerequisites that, in turn, lead to marked geographical variations.[17] Consequently, it is important to discuss each sector separately — the distribution system in this section and the production processes in the next.

Given of the need to absorb the high marketing risks and the occasional existence of law regulation, only a small number of distribution companies exist.[18] Invariably, the distribution sector

14) For example, "Spirited away" directed by MIYAZAKI Hayao won an Academy Award in 2003.
15) For instance, *op. cit.*, footnote 4) (1).
16) Scott, A. J., 'Cultural-products industries and urban economic development: prospects for growth and market contestation in global context,' *Urban Affairs Review*, 39, 2004, pp. 461–489.
17) (1) *op. cit.*, footnote 4) (3), (2) *op. cit.*, footnote 5) (1), (3) *op. cit.*, footnote 6).

dominates both publishing (or release) and production sectors through the power of sector's capital input and by controlling distribution. Thus, the cultural industry's distribution sector constitutes a "legal oligopoly."[19] Of course, the degree of oligopoly will differ between individual cultural industries and different countries. For instance, television is the primes market for Japan's animation industry; besides the ground-based broadcasting — there is a limit to radio waves — is the most prosper broadcasting way in Japan. Consequently, the industry is reliant on Japan's flagship television stations for distribution.[20] In turn, these flagship stations exercise a marked influence on the nature of production by animation firms. These uneven power relationships between the distribution and production sectors are typical of the cultural industry as a whole.

Conversely, the game industry does not have such a "strong" distribution sector. It is interesting to note that some interviewees pointed out the low profitability of the distribution sector in the game industry though the risk of excess stock is never low. This raises the question: why do companies enter such a distribution sector? The answer lies in the game industry's unique release and distribution system. The companies like Nintendo and SCE, which manufacture not only console but also media (CD-ROM, DVD-ROM etc.), are called "platform holder". As the output of the other game firms is restricted to software they are obliged to outsource under contract the fabrication of all media to a platform holder. Coupled with the consoles, the fabrication of the media is a main profit-making center for platform holders. These holders do not regard distribution sector as profitable. The only reason they participate in the distribution of products is to exercise to control over the market. By monitoring supply and demand trends and regulating the entry of software firms.[21] A platform holder confirmed during the survey that the reason for maintaining an unprofitable distribution sector was to market the consoles (interview September 2, 2002). Meanwhile, the quantity of games software sold by large software firms is so great that they have developed their own distribution sector to the save on the costs of marketing their own goods. Moreover, as revealed by the survey, they distribute not only their own products but also those from other firms to obtain economies of scale (interview August 28, 2002).[22]

Therefore, the game industry's distribution sector is not designed to benefit from its own existence. As noted, Platform holders' profits are derived from selling their consoles and manufacturing media sourced from game software firms. Thus, it is inappropriate for platform holders

18) Aksoy, A., 'Mapping the information business: integration for flexibility' (Robins, K., eds., *Understanding information: business, technology and geography*, Belhaven Press, 1992), pp. 43–60. Exactly to say, he refers to the information industry, but it is possible to think that the cultural industry comprehends almost all information industry, which he calls. Because it is consist of broadcasting industry, motion picture industry, music industry etc.

19) Despite the. "Paramount decision" — federal antitrust action — by which Hollywood major studios were forced to separate production sector and distribution sector from exhibition sector, major studios have been possessed considerable power; thus oligopoly did not actually disappear. See *op. cit.*, footnote 6)

20) By contrast, Fin / Syn Rules (Financial Interests and Syndication Rules), which existed from 1970 to 1995 in the U. S., weakened the power of three networks against the productions.

21) Fujikawa, Y., 'Sofuto kaihatsu wo suishin suru dainamizummu no gensen－Nintendo to Sony no bijinesu moderu kan kyousou－', (Takeuchi, T. et al, ed., *Mahketingu kakushin no jidai dai2kan: seihin kaihatu kakusin*, Yuhikaku, 1999), pp. 363–387. (J)

22) Platform holders had previously monopolized distribution sector, but removed of a ban on entrance of game software companies in response to their demand in the middle of 1990s (Yada, M., *Gemu rikkoku no miraizou-Sekai wo riido suru kontentsu bijinesu no subete*, NikkeiBP, 1996, p. 115. (J)).

severely to rule the game software firms. Consequently, the game software firms, which have distribution sector, cannot contract with the other game firms under over-advantageous conditions for them.

### (3) Production processes

Both the animation and game industries produce cultural products in which knowledge is embodied, but their production processes have also relatively simple and routine work (Figure 2). Many features are not dissimilar to labor-intensive manufacturing industry. Both the "original picture" in the animation industry, and "main program" and "graphic" in the game industry require a large manpower. However, the production process in the animation is much closer to manufacturing than that of the game industry, because of the difference in "reversibility". The production process in the animation is likened to a "waterfall process". Stemming originally from the software industry, this term means that the whole process is clearly divided into two parts: upstream processes and downstream processes. As the specifications and the designs are intimately defined in the upstream processes, the flow of products is one-way and the task does not return again for further reprocessing. Therefore, a multiple hierarchy subcontracting system characterizes downstream processes.[23] These explanations fit operations in the animation industry well.

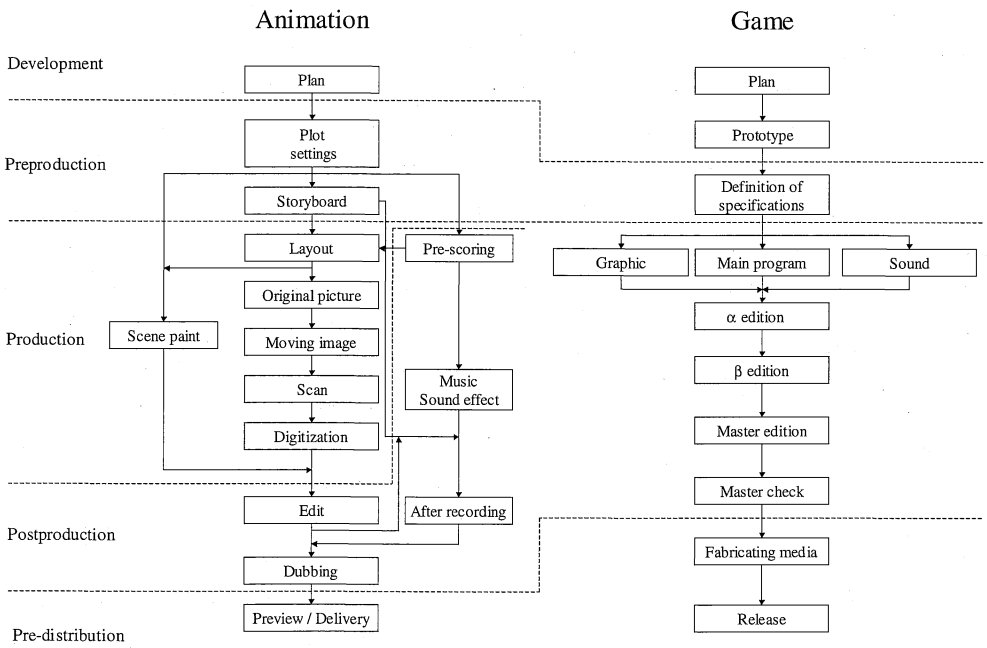



**Figure 2.** The flowchart of the animation and the home video game
source: compiled by author from
1) Nikkei BP sya gijyutsu kenkyuu bu eds., *Anime bijinesu ga kawaru*, Nikkei BP sya, 1999, p172. (J)
2) URL: http://www.meti.go/jop/policy/media_contents/down loadfiles/producer/New_Folder/3/03-17.pdf (8 August, 2004)
3) Asano, K., *Gemu gyoukai kiki ippatusu!*, Syoueisya, 2001. (J)
4) Takahashi, K., Gemu no zen shigoto 2002., Shinkigensya, 2001. (J)

"Plot and setting" and "storyboard" are processes that not only define specifications and the designs, but also the direction of the rest of processes. Within each process there are many trials and errors but radical changes about content are defined at the outset and there are no reverse flows.

Paradoxically, the game industry's production process, which is surely one of the software industries, is the antithesis of the "waterfall process". One refers to it as a "revised process"[24]. The flowchart of the game is simpler than its animation counterpart; moreover "$\alpha$ edition", "$\beta$ edition", and "master edition" are essentially the same product derived from "graphic" and "main program"[25]. In other words, these processes are intricately related to each other. This inseparability stems primarily from the obscure nature of the specifications. As a rule of thumb, the greater number of trials and errors, the better the game software output. At the outset the specification sheet is either imperfect, or unwritten. Consequently, many tasks shuttle between "graphic", "main program", and "definition of specification" stages. This difficulty in defining the intimate specifications at the outset blurs the borders between processes and the distinctiveness of preproduction, production, and postproduction stages. Thus, the game industry is essentially a knowledge industry that is less related to manufacturing, which is distinguished by progression through a set of clearly defined processes and formulaic tasks.

## III Locations

Cultural industries and cities are inseparable because the former are reliant upon external economies derived from regional agglomerations[26]. The location patterns of the animation and game industries in Japan do not deviate from this expectation. Most firms in these industries are located in Tokyo prefecture with a population over 11 million, and Osaka prefecture with a population of 8.8 million is grossly underrepresented (Table 1). Even if such deflection can be sustained by the overconcentration of manifold urban activities in Japan on Tokyo, the animation and game industries' extreme concentration in Tokyo reflects deeply rooted specific industry-related internal factors that are revealed in separate analyses of their location patterns. Although admittedly the difference is very small, the game industry has more firms outside Tokyo than the animation industry. In addition, we should pay attention to the two point about the location patterns of the game industry: first, Nintendo, one of the most famous companies in the world, is located in

**Table 1.** The number of headquarters of animation and game companies in Japan

| Prefecture | Animation | | Game | |
|---|---|---|---|---|
| | no. | % | no. | % |
| Tokyo | 220 | 80.9 | 309 | 72.4 |
| Osaka | 8 | 2.9 | 33 | 7.7 |
| Others | 44 | 16.2 | 85 | 19.9 |
| Total | 272 | 100.0 | 427 | 100.0 |

Source: See figure 3

23) Iwamoto, S. and Yoshii, H., . *"Jyouhou" no syouhinka to syouhi*, Gakubunsya, 1998. (J)

24) Sunagawa, K., 'Nihon gemu sangyou ni miru kigyousya katudou no keiki to gijyutsu senryaku: sega to namuko ni okeru sofutouea kaihatsu sosiki no keisei,' *Keisei shigaku*, 32 (4). 1998, pp. 1-27. (J)

25) "$\alpha$ edition" is first edition, with which several data — like graphics and programs — is integrated. By investigating and improving it, you finally produce "$\beta$ edition," which is nearly finished edition. Then, you improve it again and complete "master edition," which is submitted to the "platform holder."

26) *op. cit.*, footnote 3), pp. 2-15.

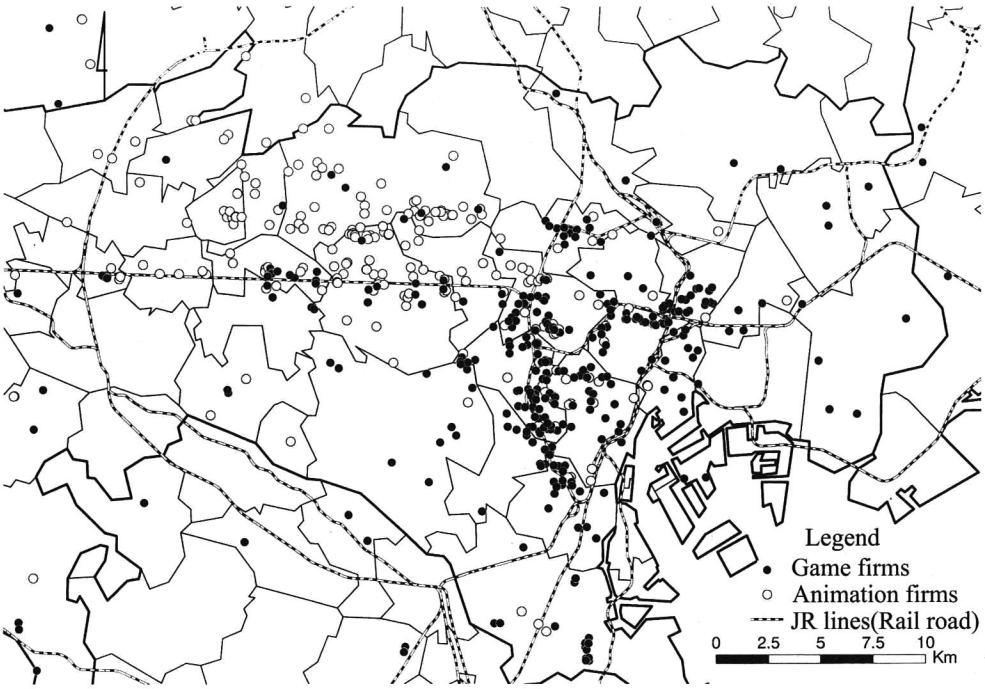



**Figure 3.** Location patterns of animation and game companies in the Tokyo region
Source:
1) http://itp.ne.jp (in June and July, 2000)
2) Risuto seisaku iinnkai., eds., *Animage anime pokketo deta 2000*, Tokuma Syotenm, 2000
3) Questionnaire survey.
4) Web site of each game company.

Kyoto prefecture adjacent to Osaka; second, it is said that the proportion of Osaka and its neighbor prefectures (Kansai) to Tokyo in game firms has become lower with the effects of Japan's deep-rooted recession on game sales since 1997.[27] Thus, regions other than Tokyo have maintained a greater presence in the game industry than in the animation industry. Within Tokyo almost all animation firms are located in the western suburbs of Tokyo whereas and the agglomeration of game firms is in the central area (Figure 3). This locational difference arises from divergences in the development paths of animation and game firms.

As most animation firms have been cloned from existing ones, their roots can be traced back to the pioneering firms: *Toei Animation* and *Mushi Production*, I should say. These firms were the two most influential ones during the early days of animation series, although *Mushi Production* was indebted to *Toei Animation* for its original staff. Many firms and workers were derived from these two companies, which were accidentally established in the western suburbs of Tokyo.[28] Hence, the offshoots are located in the same area. Another locational factor is that the animation firms need to be close to key commercial television stations, and major processing stations that are located in Tokyo.

27) Accurate statistical data does not exist, but a number of interviewees pointed to the fact.
28) *Toei Animation* was established in a part of the *Tokyo film studio* (in Ooizumi gakuen, Nerima ward) of *Toei*, whose parent company. *Mushi Production* was founded in the home of TEZUKA Osamu's own (in Fujimidai, Nerima ward).

The antecedents of games firms are more diverse. Excluding home video games, about 40% of game firms (23 of 61) in our survey entered the game industry from another industry. They were developed from: a combination of arcade games, manufacturing of amusement machines and arcade administration (6 firms); production and release of personal computer (PC) games (5 firms); system development (5 firms), production of animation or computer graphics (CG) (4 firms); other manufacturing industries (3 firms), etc. Despite their concentration within the central wards of Tokyo, this diverse set of origins has led game firms into being more dispersed than animation firms. Concentration within the game industry is related more closely to the nature of the labor market and the inter-firm relationships than the nature of their origins. Similar overriding factors make it difficult for animation firms to disperse their operations from Tokyo.

## IV Labor Markets

Workers in cultural industries are different from those in the general workforce because they are artists with a unique set of values. They include an emphasis on aesthetics, a dislike of management, neglect of monetary rewards, etc. This uniqueness is often reinforced by the specific nature of the animation and games industries.

Table 2 shows the size of firms in the animation and game industries. Both industries contain many medium and small enterprises (SMEs). However, there are more large companies in the game industry than in the animation industry. Also several game companies outstrip the largest animation company in size.[29] These differences can be partially ascribed to variations in the respective market sizes of the animation and game industries. Despite the existence of many SMEs and depressed sales, the survey revealed that, irrespective of firm size, the game industry's workforce enjoyed better employment conditions than those experienced in the animation industry. Prevailing pay levels within the animation industry are low because of the persistence of the commission system. As the price of a moving image sheet is about 200yen, the monthly incomes of many unskilled workers are under 100,000yen. Almost animation workers must start his career under such severe condition as a painter of moving images that are part of the required on-the-job training (OJT), though the pay level is improved as the development of the labor's skill.[30] Conversely, straight salary system — sometimes

**Table 2.** The size of the firms in our survey

| Parameter | Cetegory | Animation | | Game | |
|---|---|---|---|---|---|
| | | no. | % | no. | % |
| Total sales (million yen) | 0~100 | 20 | 39.2 | 13 | 23.6 |
| | 100~500 | 23 | 45.1 | 18 | 32.7 |
| | 500~1,000 | 2 | 3.9 | 2 | 3.6 |
| | 1,000~5,000 | 4 | 7,8 | 11 | 20.0 |
| | 5,000~ | 2 | 3.9 | 11 | 20.0 |
| | Total in subset | 51 | 100.0 | 55 | 100.0 |
| Employment size (both temporary and permanent) | 1~19 | 36 | 52.2 | 24 | 38.7 |
| | 20~49 | 20 | 29.0 | 12 | 19.4 |
| | 50~100 | 7 | 10.1 | 9 | 14.5 |
| | 100~ | 6 | 8.7 | 17 | 27.4 |
| | Total in subset | 69 | 100.0 | 62 | 100.0 |

Source:
1) questionnaire survey
2) interview
3) the financial reports of each firm

29) *Toei Animation* has total sales of about 16 billion yen and 260 employees in 2004 (simple body, from *Toei Animation* website), while the total sales of five of our surveyed game companies are larger than that of *Toei Animation*.

combined with payment by result — is applied to most game workers because it is difficult to distinguish between individual contributions. Moreover, working conditions are enhanced not only by the level of wages but also by the availability of insurance.

Despite the special techniques and creativity required to produce cultural products, workers in the animation and game industries are not necessarily highly educated or skilled[31] not only because are there some production processes to be simple and routine, but also because tacit knowledge and learning by doing are the prime requisite in producing cultural products.[32] OJT, therefore, prevails as the prime way of educating the worker. Invariably, a problem arises because OJT is often too burdensome for firms, notably SMEs, to carry out. Consequently, the commission system, which produces adverse conditions of employment in the animation industry, ironically eases the difficulty of new graduates in finding employment, for the animation firms does not need to pay unnecessary wages to unskilled workers. Conversely, the relative good conditions of employment burdens firms with additional costs and require them to pay a regular salary to unskilled workers. Therefore, some firms are unwilling to employ new graduates and their proportion among new recruits in the game industry is low (Table 3).

As noted, the conditions of employment, especially the respective wage structures, are quite different between the two industries. Although the relative proportion of permanent workers within each industry appears identical, there are sharp differences (Table. 4). As the treatment of temporary workers varies between firms in both industries, it is difficult to generalize. For instance, one firm regards a temporary worker as an apprentice; another firm does as a helper by individual project, and third does as a way of cost curtailment. These complications highlight the need to investigate relationships between firms and projects.

Generally, cultural products are produced as projects, which are perceived to be "temporary

**Table 3.** Proportion of new graduates in new hiring in the game industry

| Proportion of new graduates | Firms | |
|---|---|---|
| | no. | % |
| 50~100% | 20 | 39.2 |
| 0~50% | 7 | 13.7 |
| Only mid-career hiring | 12 | 23.5 |
| No new hiring | 12 | 23.5 |
| Total | 51 | 100.0 |

Source: questionnaire survey

**Table 4.** Proportion of permanet workers

| Proportion of permanent workers | Animation | | Game | |
|---|---|---|---|---|
| | no. | % | no. | % |
| 0~50% | 16 | 25.4 | 9 | 18.4 |
| 50~90% | 15 | 23.8 | 13 | 26.5 |
| 90~100% | 32 | 50.8 | 17 | 55.1 |
| Total | 63 | 100.0 | 49 | 100.0 |

Source: questionnaire survey

30) The average annual income of the animation labor is 2,470,000 yen from the questionnaire carried out — in 1999 — by "Roundtable conference thinking about the future of animation". But it is said that mid-level and skilled labors generally earn more money than general labors.

31) The vocational school provides these two industries with a lot of graduates, many interviewees, however, consider the education program of the vocational school to be inadequate and inappropriate. In the meantime, the fact that there are few colleges or universities which offer students professional education is regarded as major problem in Japanese cultural industries. Which is pointed out by Hamano, Y., 'Kokka senryaku toshiteno kontentsu bijinesu' (Zaidan houjin dejitaru kontentsu kyoukai. ed., *Dejitaru kontentsu hakusyo 2004*, Zaidan houjin dejitaru kontentsu kyoukai, 2004), pp. 10-11. (J)

32) Throsby, D., *Economics and culture*, Cambridge University Press, 2001, p. 121.

system" that require the recruitment of workers for each new task.[33] Grabher highlights that these "temporary" projects are embedded in "permanent" organizations, which means that members engaged on a new project are recruited through the personal networks cultivated both during and between projects for a long time, despite the limited-term of each project. These notions about projects can be substantiated in the animation industry, but, like the motion picture and TV program production industries, fixed inter-firm relationships cannot be ignored.[34] Conversely, game firms has "lines", which means that the constituent members of projects are sometimes fixed and unchanging; therefore the lifetime of their projects is surely short, still members of projects and the inter-firm relationships between firms are not always "temporary". By investigating the turnover and job seeking within each industry, we can comment on the degree of "permanency".

Generally, the turnover ratio is very high among cultural industries and the animation and game industries are no exceptions.[35] This turnover is not because many workers move between firms to work on the most appropriate project, as anticipated by flexible specialization theory,[36] but because of personal aversion to working conditions in certain firms ranging from disgust at low pay levels through long working hours to deterioration in human-relationships. These relationships are more potent in the game industry than in the animation industry because of the "revised" process, which requires regular and close communications between workers. Inevitably, the situation leads to intense argumentation accompanied by deep emotional conflicts. Nevertheless, both animation and game firms do not encourage high turnover ratio because it prevents them from accumulating technical capabilities. For this reason they seldom dismiss workers.

The mid-career labor market is more prosperous in the game industry than in the animation industry. Many workers to leave game firms get jobs with similar industry, while unskilled workers in the animation industry seek employment in other activities. Animation workers with higher skill levels often become freelancers or establish their own firms. Usually, workers in the game industry seek jobs through public information sources, including the websites of respective firms rather than personal networks. Headhunting is not prevalent.

To comprehend the importance of employment, we need to examine the preference of workers in seeking jobs rather than concentrating on the recruitment methods used by individual firms. Most game workers, particular those in mid-career, favor working in central Tokyo, whereas employees in the animation industry are relatively indifferent to workplace location. Why are workers in the game industry so location specific? There is no clear explanation of this phenomenon, but some interviewees commented that workers in the game industry prefer living and working in, or adjacent to, a vibrant area.

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33) (1) Grabher, G., 'Cool project, boring institutions: temporary collaboration in social context,' *Regional Studies*, 36, pp. 205–214, (2) *op. cit.*, footnote 4) (4), (3) *op. cit.*, footnote 16).

34) Regarding motion picture industry, see *op. cit.*, 4) (1), and as to TV program production industry, see *op. cit.*, 4) (4).

35) The average turnover rates in the game sector of every firm are 11.8% (34 firms). Regarding that of animation, unfortunately, there is no accurate data; nevertheless many of the interviewees of the animation firms pointed out high turnover ratio.

36) Piore, M. and Sable, C. F., *The second industrial divide: possibilities for prosperity*, Basic Books, 1984.

The unwillingness of mid-career workers to work in suburban locations has prompted game firms to seek locations in central Tokyo. Conversely, the locational tastes of workers in the animation industry are less decisive in the geographical behavior of firms. Indeed, the locational rationale of firms in the animation industry is affected more by the significance of inter-firm relationships.

## V Inter-firm Relationships

As cultural industries need largely non-renewable resources in creating each product, it is not surprising that many authors use flexible specialization theory to explain their geographical characteristics and inter-firm relationships.[37] Yet, as discussed in the preceding section, the game industry's projects are not always "temporary" and there are many fixed inter-firm relationships in both industries. Notwithstanding this observation, there is a need to assess the degree of flexible specialization within the animation industry and highlight the significance of "permanence". Then the reason for fixed inter-firm relationships can be discussed.

As animation firms are small, 34 of the 71 firms (47.9%) surveyed did not operate more than two processes (Table. 2).[38] These firms can also be distinguished by their inter-firm relationships (Table. 5). Generally, firms with preproduction processes are in the upstream group (U); those with only other processes are in the downstream group (D). Firms in Group U have more transactional linkages than Group D. Firms in Group D have fewer clients than those in Group U. Indeed, half of them have less than 5 clients. This difference stems from Group U having more opportunities to do business with a variety of companies than those in Group D, including

**Table 5.** The number of trade connections of animation firms

| | Number of clients | | | | Number of subcontractors | | | |
|---|---|---|---|---|---|---|---|---|
| | Group U | | Group D | | Group U | | Group D | |
| Number of firms | no. | % | no. | % | no. | % | no. | % |
| 0 | 2 | 5.1 | 2 | 6.9 | 3 | 9.4 | 7 | 33.3 |
| 1~4 | 15 | 38.5 | 12 | 41.4 | 5 | 15.6 | 9 | 42.9 |
| 5~9 | 10 | 25.6 | 8 | 27.6 | 5 | 15.6 | 5 | 23.8 |
| 10~29 | 9 | 23.1 | 7 | 24.1 | 14 | 43.8 | 0 | 0 |
| 30~49 | 1 | 2.6 | 0 | 0 | 5 | 15.6 | 0 | 0 |
| 50~99 | 1 | 2.6 | 0 | 0 | 2 | 6.3 | 0 | 0 |
| 100~ | 1 | 2.6 | 0 | 0 | 2 | 6.3 | 0 | 0 |
| Total | 39 | 100 | 29 | 100 | 32 | 100 | 21 | 100 |
| Aberage | 10.6 | − | 5.8 | + | 19.9 | − | 2.9 | − |

Source: Hanzawa, S., 'Tokyo niokeru animehsyon sangyou syuuseki no kouzou to henyou, *Annals of the Japan Association of Economic Geographers*, 47. 2001, pp. 288-302.

37) (1) *op. cit.*,4) (1), (2) *op. cit.*,4) (4).

38) In this case, the author divides processes into nine parts: "plan", "direction and scenario", "original picture and moving image", "scene painting", "painting", "voice and music", "filming and development", "assemble", "CG".

broadcasting stations, publishers, and game firms. Conversely, firms in Group U have more subcontractors than Group D firms. These results from principal contractors within Group U have to draw upon other firms to complete the animation as they have only a limited range of processes.

Vertical disintegration and multi-layered transaction relations can be traced to the beginnings of animated television broadcasts during the 1960s. Animated TV series covers various kinds of stories and pictures and only needs a very short production period to meet weekly deadlines. This schedule makes it difficult to predict market trends. As noted, the quality of external production was expected to lower than that of in-house production. However, animated television series did not require the high quality output of cinema, which previously was the main market for animation prior to television. Furthermore, the high rate of percentage pay in those days prompted animation workers to become freelances.[39] Therefore, vertical disintegration developed, and animation firms and freelances must be adjacent to each other, so as to maximize the benefit of inter-corporate goods circulation and information exchange.

These findings suggest that all inter-firm relationships are flexible. However, permanent business connections persist and play an important role in the production of animation. This is illustrated by an interviewer's response:

> Our subcontractors are not fixed, but we intend to produce animation products, which are scored 70−80 points, constantly as our company focuses on TV series animation; therefore, we reserve stable drawing teams, filming teams, and painting teams. They maintain 40% of our production capability. They are collaborators rather than subcontractors.[40]

Actually, animation firms find it easier to deal with fixed and familiar inter-firm relationships than with new and unfamiliar ones. Already, animation firms have assessed their reliability and capability. Nevertheless, animation firms seek to counterbalance their fixed relationships with flexible arrangements.

Inter-firm relationships within the game industry are subdivided into two types: "whole-transactions" and "partial-transactions." Whole transactions imply that the total production process is outsourced; partial transactions mean that limited part of the process is outsourced. Generally, most firms prefer in-house production to outsourcing the entire production since creators want to produce their own products by themselves. However, outsourcing the whole transaction enables them to overcome the stumbling block of releasing new genre and conserve their own production resources, particularly where they have a stock of new ideas and popular characters. Therefore, whole transaction prevails as a means of coping with the prolonged recession in Japan. Although the whole-transaction involves switching large funds from outsourcer to subcontractor,[41] compared with the partial-transaction, there are few risks because only a limited number of orders are involved and the recipient is chosen on the basis of past reliability. As close and

39) In the animation industry, we may as well often regard the freelance as a firm because a lot of animators work in their own home as one of subcontractors.

40) From the interview with one of major production companies (on November 1, 2000).

continuing information exchange is unnecessary both outsourcers and subcontractors, the distance between them is immaterial. On the contrary, the necessity of intimate face-to-face communication in partial-transactions due to the ambiguity of specifications leads outsourcers and subcontractors to being located in close proximity to each other, and also makes it difficult to do outsourcing. Consequently, compared with the animation industry, the partial-transaction is not active and absolute.

Regrettably, the author does not possess accurate information on the number of transactions between game firms. However, an investigation of the inter-firm relationships among leading firms reveal a variety of arrangements.[42] For example, one firm is outsourcing the whole transaction to 20 subcontractors and, on occasions, outsources a partial process to one of them with which the company has dealt for 20 years; another firm boasts no long-term relationships with other firms because in-house production is fundamental and subcontracting is restricted to either voices or computer graphics production; a third firm has a regular arrangement with 10 subcontractors, three of which have been associated with the company for many years; and the last firm has between six and ten partial-subcontractors, most of which have been engaged in graphics, but it has recently commenced to outsource the whole of the production processes. These observations suggest that: these leading firms' subcontractors are not many despite their largeness (cf. Table 5); the number of their subcontractors are surely larger than that of most game firms; and most contractors, with the exception of the third firm, have long-term relationships with subcontractors. Moreover, game firms prevent primary subcontractors from outsourcing to secondary subcontractors for fear of losing control of proprietary material. Hence, the transactional system of relations within the game industry is single-layered contrary to the multi-layered pattern exhibited by the animation industry. Consequently, a firm's project members — subcontractors and employees — are fixed. At best, only a few firms are involved with a single project. Thus, most game firms have fewer, but more permanent, inter-firm relationships compared with the animation industry.

Examining the limited number of inter-firm relationships from another perspective, we discover that there is an abundance of the firms engaged in the release sector. According to our survey, 40 of the 56 firms had released at least one game software package between 1999 and 2001. To release in the game industry is to produce or make subcontractors develop the game software depending on their own funds,[43] and to possess property rights. In contrast, only a few animation firms have the financial capacity to raise production funds and a reliant on flagship television stations to provide the wherewithal in return for all property rights. Concerning the television series animation, animation firms always have at least one client — the TV broadcasting station — and are always just subcontractors; while not all game firms always have clients. Because they are much more independent of other firms than their counterparts in animation. As noted,

41) The standard production budget of PlayStation2 software is about 2–3 hundred million yen.

42) In 2002, top 10 firms occupied about 75 % of Japanese domestic market share. The author interviewed 4 of the top 10 firms.

43) In Japan, almost production fund is self-financed. No bank, no investor, and no other company invest game software with minor exception.

this difference stems from the existence of a "platform holder"

## VI Conclusion

We have contrasted two Japanese cultural industries — animation and home video games — by examining their key characteristics: location patterns, labor markets, and inter-firm relationships. These key factors are not only interrelated but also create two distinctive industrial geographies. Both industries are concentrated in small firms within Greater Tokyo and both exhibit close inter-firm relationships, though most animation firms are located in western suburbs, whereas most game ones are in the central area. It is caused by the difference between these industries in the degree of connections with the firms of similar industry. Most animation firms have been cloned from existing ones and must deal with a lot of the other animation firms for lack of some (or most) production processes within themselves. Concerning game firms, their roots are ascribed to the various industries, besides they can produce one game software product with no firm or only a few ones because of the difficulty in subdividing their processes into each part. Therefore, the strength of animation inter-firm "bonds" makes those firms agglomerate around the birthplace of the industry — *Toei Animation* and *Mushi Production* —, while the weakness of game ones do not always leads to the free location decisions of game firms because of the mid-careers' preference to working in central Tokyo.

Observing their inter-firm relationships only in detail, it is interesting to note that they are, not only implicitly but also sometimes explicitly, permanent. In this point of view, game firms have fewer inter-firm relationships and less flexibility in changing them than animation firms. Nevertheless, each firm within the Japanese game industry is free to manufacture its product without interference from the distribution sector. Conversely, production in the Japanese animation industry is dominated by a strong distribution sector. Therefore, the sharp differences between the two industries stem from their peculiar distribution systems — the existence of the TV flagship stations' "legal oligopoly" in the animation industry and "platform holders" in the game industry and — and production processes — "waterfall process" in the former and "revised process" in the latter —, which influence each other and the behaviors of their component firms.

Based on the findings, the overriding feature of Japanese cultural industries is the absence of any institutional interference. There is no official governmental policy, no labor union, and poor supportive educational organizations. Japanese cultural industries have developed entirely under market rule or *laisser-faire*; in other words any prosperity they have enjoyed cannot be attributed to institutional interference. This is quite different from cultural industries in other parts of the world. However, the Japanese government has undergone a recent change in its position and now recognizes the critical importance of content[44]: training producers, strengthening the protection of intellectual property, supporting trade fairs, etc. Government intervention will need to be monitored by scholars in tracing the ongoing transformation of Japanese cultural industries.

44) In Japan, the term "content industry" is popular and substitutes for "cultural industry". "Content promotion law (short title)" was approved in May 2004.

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# The Japanese Animation and Home Video Game Industries: Locational Patterns, Labor Markets, and Inter-firm Relationships.

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The animation and home video game industries represent two of the most widely-known contemporary Japanese cultural industries. Both industries share similar characteristics with high turnover ratio and locational concentration in Tokyo. However, they differ markedly in their detailed location patterns, labor markets, and inter-firm relationships. The distribution of animation firms is more concentrated on the national scale in Tokyo and in the western suburbs of Tokyo on the local scale than that of game ones. Workers in the game industry are employed through public advertisements, whether recruiting new graduates or mid-career staff, and they often move one firm to another because there is occasional serious deterioration in human relationships. Conversely, in the animation industry there is little apparent deterioration in human relationships. Most job leavers of the animation become freelancers, establish their own firms or leave the industry entirely. Game firms have fewer inter-firm relationships and less flexibility to alter their business partners than animation firms.

These differences stem from their peculiar distribution systems — the existence of the "legal oligopolistic" TV flagship stations of in the animation industry and "platform holders" in the game industry — and production processes — "waterfall process" in the former and "revised process" in the latter —, which not only influence each other at an industry level but also the behaviors of their individual component firms.

**Key words**: animation industry, home video game industry, agglomeration, labor market, inter-firm relationship

# 日本のアニメーション産業と家庭用ビデオゲーム産業
## ——立地・労働市場・企業間関係——

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アニメーション産業と家庭用ビデオゲーム産業は，現代日本の最も広く知られた文化産業のうち2つである。両産業は，高い離職率と東京への集中という共通の特徴を有する。しかし，詳細な立地や，労働市場，企業間関係という点では明らかに違いがある。ゲーム会社に比べてアニメーション会社の立地は，国単位ではより東京に，地域単位ではより東京西部に集中する。ゲーム産業の労働者は，新卒であろうと中途であろうと公募を通じて採用され，時折深刻な人間関係の悪化があるためしばしば企業間を移動する。逆に，アニメ産業では，明らかな人間関係の悪化は少ない。アニメの離職者のほとんどは，フリーランサーになるか自分の会社を設立するかして，そうでなければ完全に当該産業から離れる。ゲーム会社は，アニメ会社に比べ取引関係が少なく，取引先を替える柔軟性も小さい。

これらの違いは，特有の流通システム——アニメ産業における「合法寡占的」テレビキー局と，ゲーム産業におけるプラットフォームホルダーの存在——と特有の制作工程——前者の「ウォーターフォール工程」と後者の「リバイズド工程」——から生じる。それらは，産業レベルで相互に影響するだけではなく，個々の企業の行動にも影響を与える。

キーワード：アニメーション産業，家庭用ビデオゲーム産業，集積，労働市場，企業間関係